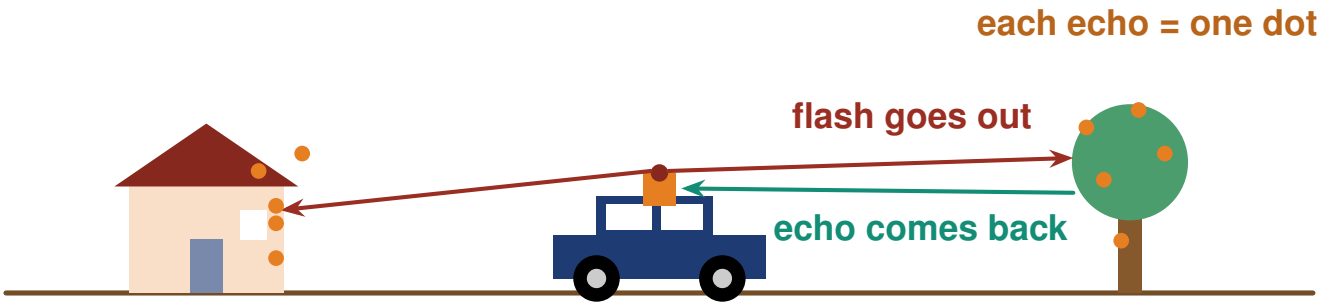
What Is LiDAR?

BIG IDEA

LiDAR measures how far away things are by timing flashes of light, just like an echo.



THE IDEA

Clap your hands in a big gym and you hear an echo. A far wall sends the echo back later than a near one. A LiDAR does the same thing with tiny flashes of laser light, hundreds of thousands of times every second. Each echo becomes one dot. Millions of dots together make a 3-D picture called a **point cloud**.

REMEMBER

- ✓ The flash goes out, bounces off something, and comes back.
- ✓ Every echo becomes one dot. Lots of dots make a **point cloud**.
- ✓ On a moving car, we must also know where the scanner was and which way it faced for every flash.

TRY THIS

In a gym or a big hallway, clap once and listen for the echo. Stand closer to the wall and clap again. Does the echo come back faster?